KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

YEAR 10 TRANSITION

EXTENSION BOOKLET

FOR FAST FINISHERS · CHALLENGE QUESTIONS

CHAPTER 4

YEAR 10 MATHEMATICS · CHAPTER 4 · EXTENSION: FAST FINISHERS

Probability and statistics

One page of harder questions for each exercise in Chapter 4. Finish the exercise in your Chapter 4 booklet first, then come here. Worked answers are at the back.

EXERCISE	EXTENSION TOPIC	PAGE	DONE
4A	Review of probability	2	☐
4B	Venn diagrams and two-way tables	3	☐
4C	Using relative frequencies to estimate probabilities	4	☐
4D	Types of data and collecting data	5	☐
4E	Displaying and describing data	6	☐
4F	Frequency tables and histograms	7	☐
4G	Time-series data	8	☐
4H	Measures of centre and spread	9	☐

EVERY CARD HAS A TYPE

Understanding
Reasoning
Problem solving

MIXED IN

Review 4A questions from earlier exercises

NO DIAGRAM?

✎ Draw a diagram first, in the dotted space

CHALLENGE

★☆☆ to ★★★

The last card on each page is the challenge.

NAME _____ CLASS _____ TEACHER _____

Artwork: a tiny copy of the whole set, hanging on branching filaments, part of the Mandelbrot set, coloured gold.

Review of probability

FAST FINISHERS

EXTENSION Finished exercise 4A? Try these. Show your working in each box. Write probabilities as fractions in simplest form.

Understanding | Reasoning | Problem solving

1 Understanding ★☆☆

This spinner has 8 equal sections: R red, B blue, G green, Y yellow.



a P(red)
b P(not blue)
c P(red or yellow)
d P(purple)

a _____ b _____ c _____ d _____

2 Understanding ★☆☆

A letter is chosen at random from the word **MATHEMATICS**.

a How many letters are there?
b Find P(M).
c Find P(a vowel).
d Find P(not T).

a _____ b _____ c _____ d _____

3 Reasoning ★★☆

Find the error. Josh says, "When you roll a die you either get a 6 or you don't. So P(6) = ½."

a Explain why Josh is wrong.
b Find the correct probability of rolling a 6.

b _____

4 Problem solving ✎ Draw a diagram ★★☆

Work backwards. A bag has 5 red, 3 blue and some green counters. The probability of red is ½.

a How many counters are in the bag?
b How many are green?
c Find P(blue).

a _____ b _____ c _____

5 Understanding ★★☆

Two coins are tossed.

a List the sample space.
b Find P(two heads).
c Find P(at least one head).
d Find P(one head and one tail).

b _____ c _____ d _____

6 Reasoning ★★★

Two dice are rolled and the numbers are added.

a Complete the table of totals.
b Find P(total of 7).
c Find P(total more than 9).
d Which total is most likely? Explain.

	1	2	3	4	5	6
1	2	3	4	5	6	7
2						
3						
4						
5						
6						

b _____ c _____ d _____

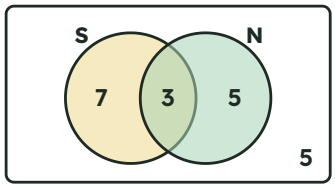
Venn diagrams and two-way tables

FAST FINISHERS

EXTENSION Finished exercise 4B? Try these. Show your working in each box. Check that the numbers in your diagram or table add to the total. Understanding · Reasoning · Problem solving

1 Understanding ★☆☆

20 students were asked if they play soccer (S) or netball (N).



a How many play soccer?
b Find P(both).
c Find P(neither).
d Find P(netball only).

a _____ b _____ c _____ d _____

2 Understanding ★☆☆

30 families were asked if they own a cat or a dog. Copy and complete the table.

	Dog	No dog	Total
Cat	4	6	10
No cat	9		
Total			30

b Find P(cat and dog).
c Find P(no pets).

b _____ c _____

3 Reasoning ✎ Draw a diagram ★★☆

Of 30 students, 18 like maths (M), 14 like music (Mu) and 6 like both.

a Draw a Venn diagram showing this.
b How many like neither?
c Find P(maths only).

b _____ c _____

4 Problem solving ★★☆

40 people were surveyed. 22 drink coffee, 15 drink tea and 8 drink both.

	Coffee	No coffee	Total
Tea			
No tea			
Total			40

b Find P(neither).
c Find P(tea or coffee).

b _____ c _____

5 Reasoning Review 4A ★★☆

Events are **mutually exclusive** if they cannot happen at the same time. Are these mutually exclusive? Explain.

a Rolling a die: getting a 2 and getting an odd number.
b Choosing a card from a deck: getting a heart and getting a king.
c Find P(heart or king) for a 52-card deck.

a _____ b _____ c _____

6 Problem solving ✎ Draw a diagram ★★★

Work backwards. In a class of 28, 17 study Art, 12 study Drama and 4 study neither.

a How many study both? Explain your method.
b Draw a Venn diagram.
c Find P(a student studies exactly one of the subjects).

a _____ c _____

Using relative frequencies to estimate probabilities

FAST FINISHERS

EXTENSION Finished exercise 4C? Try these. Show your working in each box. Relative frequency = frequency ÷ total number of trials.

Understanding · Reasoning · Problem solving

1 Understanding ★☆☆

A coin is tossed 50 times and lands on heads 28 times.

a Find the relative frequency of heads.

b How many heads would you expect in 50 tosses?

c Does this mean the coin is unfair? Explain.

a _____ b _____

2 Understanding ★☆☆

A die is rolled 60 times.

1	2	3	4	5	6
8	11	9	12	7	13

a Find the relative frequency of a 6.

b Find the relative frequency of an even number.

c How many 6s would you expect in 60 rolls?

a _____ b _____ c _____

3 Reasoning ★★☆

Mel counts 200 cars passing her school. 46 are white.

a Estimate the probability that the next car is white.

b How many white cars would you expect in the next 1000 cars?

a _____ b _____

4 Problem solving ★★☆

Mia shoots 48 free throws and scores 36.

a Estimate the probability that she scores.

b How many would you expect her to score in her next 20 shots?

c Would 200 shots give a better estimate? Explain.

a _____ b _____

5 Reasoning ★★☆

A spinner with three colours is spun 200 times.

Red	Blue	Green
42	38	120

a Do you think the three colours have equal sections? Explain.

b Estimate what fraction of the spinner is green.

b _____

6 Problem solving Review 4B ★★★

80 people were asked how they mostly watch TV shows.

	Streaming	Free-to-air	Total
Under 30	28	7	
30 and over	20	25	
Total			80

a Complete the totals.

b Estimate the probability that a person prefers streaming.

c Estimate P(streaming) for people under 30.

d In a town of 5000 people, how many would you expect to prefer streaming?

b _____ c _____ d _____

Types of data and collecting data

EXTENSION Finished exercise 4D? Try these. Show your working in each box. Categorical: nominal or ordinal. Numerical: discrete or continuous. Understanding · Reasoning · Problem solving

1 Understanding ★☆☆

Classify each type of data.

a the number of siblings you have
b your height
c your favourite colour
d shirt sizes S, M, L, XL
e postcodes (e.g. 2487)

2 Understanding ★☆☆

A **census** collects data from everyone. A **sample** uses part of the group.

a Recording the height of every student in your class: census or sample?
b Testing 50 light globes from a batch of 10 000: census or sample?
c Why would the factory not test every globe?

a _____ b _____

3 Reasoning ★★☆

A council wants to know how residents travel to work. It surveys people at the train station at 8 am.

a Explain why the results may be biased.
b Suggest a better way to collect the data.

4 Reasoning ★★☆

A survey asks: *"Don't you agree that school should start later?"*

a Explain why this is a poor survey question.
b Rewrite the question so that it is fair.

5 Problem solving ★★☆

20 students were asked how many pets they have:

0 2 1 1 3 0 2 1 4 1 0 2 1 3 1 2 0 1 2 1

a Make a frequency table.
b What is the mode?
c What percentage of students have 2 or more pets?

b _____ c _____

6 Problem solving ★★★

A school of 900 students wants a sample of 60 that represents each year group fairly.

Yr 7	Yr 8	Yr 9	Yr 10	Yr 11	Yr 12
180	165	150	150	135	120

a What fraction of the school is in the sample?
b How many from each year should be in the sample?

a _____

Displaying and describing data

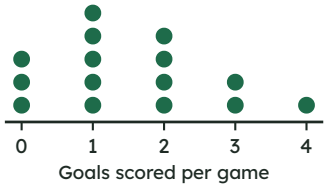
FAST FINISHERS

EXTENSION Finished exercise 4E? Try these. Show your working in each box. For stem-and-leaf plots, always read the key.

Understanding · Reasoning · Problem solving

1 Understanding ★☆☆

The dot plot shows the goals a team scored in each game.



a How many games were played?
b What is the mode?
c In what percentage of games did they score 3 or more?

a _____ b _____ c _____

2 Understanding ★☆☆

a How many values are there?
b Find the range.
c Find the mode.
d Find the median.

Stem	Leaf
1	5 8
2	0 3 3 7
3	1 4 6 9
4	2 5

2 | 3 means 23

a _____ b _____ c _____ d _____

3 Reasoning ★★☆

Scores in a quiz:

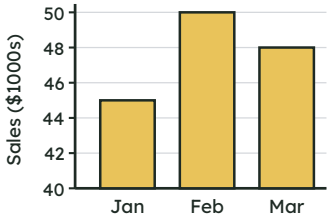
34 27 41 29 36 22 38 45 31 27 33 40

a Draw an ordered stem-and-leaf plot with a key.
b Find the median.
c Is the data roughly symmetrical or skewed?

b _____ c _____

4 Reasoning ★★☆

A shop shows its sales on this graph.



a Why does the graph make February look much better than January?
b Find the real percentage increase from January to February.

b _____

5 Problem solving ★★☆

Test scores for two classes.

a Find the median of each class.
b Find the range of each class.
c Which class did better? Give a reason.

Class A	Stem	Class B
	4	5 9
8 2	5	5
7 4 1	6	0 2 6 8
5 3 0	7	1 9
8 1	8	5

6 | 1 means 61

A _____ B _____ c _____

6 Reasoning Review 4D ★★★

This plot shows push-ups done in one minute. One leaf is missing. The range is 26.

Stem	Leaf
1	2 5 8
2	1 4 4 7
3	0 □

2 | 1 means 21 push-ups

a Find the missing leaf.
b Find the median.
c Is this data discrete or continuous? Explain.

a _____ b _____ c _____

Frequency tables and histograms

FAST FINISHERS

EXTENSION Finished exercise 4F? Try these. Show your working in each box.
Percentage frequency = frequency ÷ total × 100%.

Understanding · Reasoning · Problem solving

1 Understanding ★☆☆

Copy and complete the table.

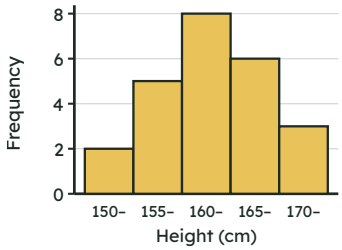
Class	Frequency	% frequency
0–9	3	
10–19	7	
20–29	6	
30–39	4	
Total		

b Which is the modal class?

b _____

2 Understanding ★☆☆

The histogram shows the heights of a group of students.



a How many students are there?
b Which is the modal class?
c How many students are 165 cm or taller?
d What percentage are under 160 cm?

a _____ b _____ c _____ d _____

3 Reasoning ★★☆

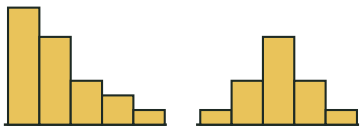
Minutes 20 students spent on their phone before school:

12 25 7 33 18 41 22 15 9 28 36 19 24 31 14 27 20 38 11 26

a Make a frequency table using classes 0–9, 10–19, … 40–49. Include a % frequency column.
b Describe the shape of the data.

4 Reasoning ★★☆

a Which graph is symmetrical and which is positively skewed?
b Give a real example of data that might look like graph A.



a _____

5 Problem solving Review 4E ★★☆

a Make a frequency table with classes 10–19, 20–29, 30–39 and 40–49.
b Which is the modal class?
c What percentage of values are 30 or more?

Stem	Leaf
1	3 6 9
2	0 2 5 5 8
3	1 4 7
4	0

2 | 5 means 25

b _____ c _____

6 Problem solving ★★★

Work backwards. 40 students did a quiz out of 19.

Score	% freq.	Frequency
0–4	10%	
5–9		
10–14	30%	
15–19	20%	

a Find the missing percentage.
b Find each frequency.
c How many students scored 10 or more?
d Can you tell how many scored exactly 12? Explain.

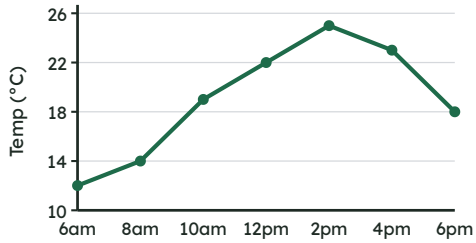
a _____ c _____

Time-series data

FAST FINISHERS

EXTENSION Finished exercise 4G? Try these. Show your working in each box. A time series shows how data changes over time. Look for a trend. Understanding | Reasoning | Problem solving

1 Understanding ★☆☆



a What was the highest temperature, and when?
b By how much did it rise from 6 am to 2 pm?
c In which 2-hour period did it rise the fastest?

a _____ b _____ c _____

2 Understanding ★☆☆

Electric cars sold in a state (thousands):

2019	2020	2021	2022	2023	2024	2025
2	3	5	9	14	21	30

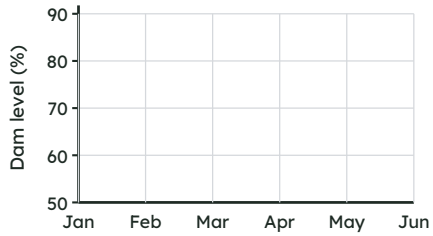
a Describe the trend.
b Is the number growing by the same amount each year? Explain.
c Predict the sales for 2026.

c _____

3 Reasoning ★★☆

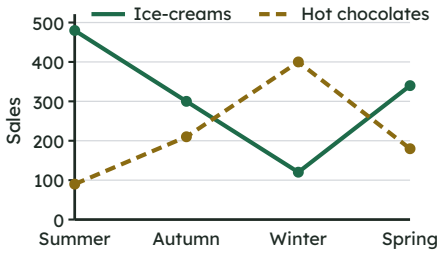
Dam levels: Jan 82%, Feb 76%, Mar 70%, Apr 65%, May 61%, Jun 58%.

a Plot the data on the grid and join the points.
b Describe the trend.
c Predict the level for July.



c _____

4 Problem solving ★★☆



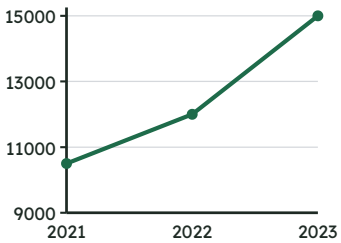
a In which season were the most hot chocolates sold?
b Describe the pattern in the two graphs.
c In which season did hot chocolate outsell ice-cream?

a _____ c _____

5 Reasoning Review 1A ★★☆

The graph shows visitors to a museum.

Find the error. Tim says, "Visitors went up 100% from 2022 to 2023, because the graph doubled in height."



a Explain Tim's mistake.
b Find the real percentage increase.

b _____

6 Problem solving ✎ Draw a diagram ★★★

Population of a town:

2000	2005	2010	2015	2020	2025
4200	4600	5100	5500	6000	6400

a Draw a time-series graph.
b Find the average increase every 5 years.
c Predict the population in 2030.
d Why are predictions far into the future less reliable?

b _____ c _____

Measures of centre and spread

FAST FINISHERS

EXTENSION Finished exercise 4H? Try these. Show your working in each box. Mean = sum ÷ number of values. Median = middle value in order. Understanding · Reasoning · Problem solving

1 Understanding ★☆☆

a For 4, 7, 7, 9, 13 find the mean, median, mode and range.

b For 12, 15, 18, 20, 23, 26 find the median.

mean _____

median _____

mode _____

range _____

b median _____

2 Understanding ★☆☆

Goals scored in 20 games:

Goals	0	1	2	3	4
Games	4	6	5	3	2

a Find the total number of goals.

b Find the mean.

c Find the mode.

d Find the median.

a _____ **b** _____ **c** _____ **d** _____

3 Reasoning ★★☆

The ages of people in a car pool are 14, 15, 15, 16, 15, 14, 16 and 62.

a Find the mean age.

b Find the median age.

c Which better describes a typical age? Explain.

a _____ **b** _____

4 Problem solving ★★☆

Work backwards. The mean of 5 test scores is 72. Four of the scores are 65, 80, 70 and 74.

a Find the total of all 5 scores.

b Find the fifth score.

a _____ **b** _____

5 Reasoning Review 4E ★★☆

a Find the mean.

b Find the median.

c Find the mode.

d Find the range.

Stem	Leaf
2	3 5 8
3	0 4 4 9
4	1 6

3 | 4 means 34

a _____ **b** _____ **c** _____ **d** _____

6 Problem solving ★★★

Ben's mean score for 4 tests is 68. Each test is out of 100.

a What must he score on test 5 to raise his mean to 72?

b Can he raise his mean to 85 with one more test? Explain.

c What is the highest mean he could have after 5 tests?

a _____ **c** _____

ANSWERS Worked answers. Check your method as well as your answer. Rounded answers may differ slightly in the last decimal place.

4A · Review of probability

1. **a** $^3/_8$ **b** $^6/_8 = ^3/_4$ **c** $^4/_8 = ^1/_2$ **d** **0** (impossible)
2. **a** **11** **b** $^2/_{11}$ **c** A, E, A, I: $^4/_{11}$ **d** $^9/_{11}$
3. **a** The two outcomes are not equally likely: "not 6" covers five outcomes (1–5). **b** $^1/_6$
4. **a** 5 is half the total, so **10** **b** 10 − 5 − 3 = **2** **c** $^3/_{10}$
5. **a** **HH, HT, TH, TT** **b** $^1/_4$ **c** $^3/_4$ **d** $^2/_4 = ^1/_2$
6. **a** 36 totals from 2 to 12 **b** $^6/_{36} = ^1/_6$ **c** 10, 11, 12: 3 + 2 + 1 = 6, so $^1/_6$ **d** **7**; it appears 6 times, more than any other total.

4B · Venn diagrams and two-way tables

1. **a** 7 + 3 = **10** **b** $^3/_{20}$ **c** $^5/_{20} = ^1/_4$ **d** $^5/_{20} = ^1/_4$
2. **a** No cat/no dog **11**; no cat total **20**; dog total **13**; no dog total **17** **b** $^4/_{30} = ^2/_{15}$ **c** $^{11}/_{30}$
3. **a** M only 12, both 6, Mu only 8, neither 4 **b** 30 − 12 − 6 − 8 = **4** **c** $^{12}/_{30} = ^2/_5$
4. **a** Tea: 8, 7, 15. No tea: 14, 11, 25. Total: 22, 18, 40 **b** $^{11}/_{40}$ **c** 8 + 7 + 14 = 29, so $^{29}/_{40}$
5. **a** **Yes**: 2 is not odd. **b** **No**: the king of hearts is both. **c** 13 + 4 − 1 = 16, so $^{16}/_{52} = ^4/_{13}$
6. **a** 28 − 4 = 24 study at least one; 17 + 12 = 29, so **5** are counted twice. **b** Art only 12, both 5, Drama only 7, neither 4 **c** $^{19}/_{28}$

4C · Using relative frequencies to estimate probabilities

1. **a** 28 ÷ 50 = **0.56** **b** **25** **c** Not necessarily. Results vary by chance; 28 is close to 25.
2. **a** $^{13}/_{60} \approx$ **0.22** **b** (11 + 12 + 13) ÷ 60 = **0.6** **c** $^1/_6 \times 60$ = **10**
3. **a** 46 ÷ 200 = **0.23** **b** 0.23 × 1000 = **230**
4. **a** 36 ÷ 48 = **0.75** **b** 0.75 × 20 = **15** **c** **Yes**. More trials give a more reliable estimate.
5. **a** **No**. With equal sections each colour would come up about 67 times, but green came up 120 times. **b** 120 ÷ 200 = $^3/_5$ (about 60%)
6. **a** 35, 45; 48, 32 **b** 48 ÷ 80 = **0.6** **c** 28 ÷ 35 = **0.8** **d** 0.6 × 5000 = **3000**

4D · Types of data and collecting data

1. **a** **numerical, discrete** **b** **numerical, continuous** **c** **categorical, nominal** **d** **categorical, ordinal** **e** **categorical, nominal**: postcodes are labels, not amounts.
2. **a** **census** **b** **sample** **c** Testing destroys the globes, and it would take too long and cost too much.
3. **a** Almost everyone there uses the train, so drivers, cyclists and people who work from home are missed. **b** Survey a random sample of residents, e.g. by post or online, at different places and times.
4. **a** It is a leading question that pushes people to say yes. **b** E.g. "What time would you like school to start?" or "Should school start earlier, later or at the same time?"
5. **a** 0: 4, 1: 8, 2: 5, 3: 2, 4: 1 (total 20) **b** **1** **c** (5 + 2 + 1) ÷ 20 = **40%**
6. **a** $^{60}/_{900} = ^1/_{15}$ **b** ÷ 15: **12, 11, 10, 10, 9, 8** (total 60)

ANSWERS Worked answers. Check your method as well as your answer. Rounded answers may differ slightly in the last decimal place.

4E · Displaying and describing data

1. **a** 3 + 5 + 4 + 2 + 1 = **15** **b** **1 goal** **c** 3 ÷ 15 = **20%**
2. **a** **12** **b** 45 − 15 = **30** **c** **23** **d** (27 + 31) ÷ 2 = **29**
3. **a** 2 | 2 7 7 9; 3 | 1 3 4 6 8; 4 | 0 1 5 **b** (33 + 34) ÷ 2 = **33.5** **c** **roughly symmetrical**
4. **a** The vertical axis starts at 40, not 0, so the Feb column looks twice as tall. **b** 5 ÷ 45 = **11.1%**
5. **a** A: (67 + 70) ÷ 2 = **68.5**; B: (62 + 66) ÷ 2 = **64** **b** A: 88 − 52 = **36**; B: 85 − 45 = **40** **c** **Class A**: higher median and less spread.
6. **a** 12 + 26 = 38, so the leaf is **8** **b** 9 values, so the 5th: **24** **c** **Discrete**: push-ups are counted in whole numbers.

4F · Frequency tables and histograms

1. **a** Total 20; 15%, 35%, 30%, 20% (100%) **b** **10–19**
2. **a** 2 + 5 + 8 + 6 + 3 = **24** **b** **160–164 cm** **c** 6 + 3 = **9** **d** 7 ÷ 24 = **29.2%**
3. **a** 2 (10%), 6 (30%), 7 (35%), 4 (20%), 1 (5%) **b** **roughly symmetrical**, peaking at 20–29 minutes
4. **a** **B is symmetrical**; **A is positively skewed** (tail to the right). **b** E.g. the number of pets per family, or weekly income: most values are small and a few are large.
5. **a** 3, 5, 3, 1 (total 12) **b** **20–29** **c** 4 ÷ 12 = **33.3%**
6. **a** 100 − 60 = **40%** **b** **4, 16, 12, 8** **c** 12 + 8 = **20** **d** **No**. Grouped data only shows how many are in 10–14, not each score.

4G · Time-series data

1. **a** **25 °C at 2 pm** **b** 25 − 12 = **13 °C** **c** **8 am to 10 am** (up 5 °C)
2. **a** **Upward** trend **b** **No**: the increases are 1, 2, 4, 5, 7, 9, so it is growing faster each year. **c** About **40 000** (anything about 38 000–45 000 with a reason)
3. **a** Points plotted and joined with straight lines **b** **Downward**, but falling more slowly each month **c** About **55%**
4. **a** **Winter** **b** Seasonal and opposite: ice-cream is high in summer and low in winter; hot chocolate is the reverse. **c** **Winter** only
5. **a** The axis starts at 9000, not 0, so heights above the axis are misleading. **b** 3000 ÷ 12 000 = **25%**
6. **a** Graph with year on the horizontal axis **b** (6400 − 4200) ÷ 5 = **440** **c** 6400 + 440 ≈ **6840** **d** The trend may change (new housing, jobs, people moving away).

4H · Measures of centre and spread

1. **a** mean 40 ÷ 5 = **8**; median **7**; mode **7**; range **9** **b** (18 + 20) ÷ 2 = **19**
2. **a** 0 + 6 + 10 + 9 + 8 = **33** **b** 33 ÷ 20 = **1.65** **c** **1** **d** 10th value is 1, 11th is 2, so **1.5**
3. **a** 167 ÷ 8 = **20.9** **b** **15** **c** The **median**: the outlier (62) pulls the mean up.
4. **a** 5 × 72 = **360** **b** 360 − 289 = **71**
5. **a** 300 ÷ 9 = **33.3** **b** **34** **c** **34** **d** 46 − 23 = **23**
6. **a** 5 × 72 − 4 × 68 = 360 − 272 = **88** **b** **No**: he would need 425 − 272 = 153, more than 100. **c** (272 + 100) ÷ 5 = **74.4**

ABOUT THE ARTWORK

The pictures in this booklet are parts of the Mandelbrot set, drawn by a computer that repeats one simple rule, $z \to z^2 + c$, hundreds of times for every point. This one shows a tiny copy of the whole set hanging on branching filaments. Zoom in on any part of the edge and you find more detail, forever. Maths can make beautiful things.

ACKNOWLEDGEMENT OF COUNTRY

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.